

Hey, everybody. Welcome to your chapter 8 review. In this episode, we're going to cover exercise metabolism and bioenergetics. Personally, I've seen or heard about half a dozen ads for supplements that boost your metabolism just this week. There's a lot of science that goes into truly understanding how metabolism works. And it's going to be important to comprehend it in order to effectively support your clients' goals. We all know the basic premise of energy intake versus energy output and how that affects our bodies. In this chapter, we'll take a deeper dive to gain a true understanding of the chemical processes and their relation to health and fitness.

By the end of the chapter, you need to be able to summarize how the first law of thermodynamics governs energy usage, describe the energy systems of the human body, identify how each macronutrient can be used as a fuel source, integrate the concepts of energy balance in relation to body composition management, and identify how to efficiently fuel the body in relation to activity intensity.

Our bodies need a constant supply of energy to function properly, maintain health, and support physical activity. Exercise places unique demands on the energy production pathways. Food is what provides our cells with the energy to survive. Before food can become a usable form of energy, it has to be converted into smaller units called substrates, including carbohydrates, proteins, and fats. These substrate molecules are then converted into ATP, which is the energy currency of all cells.

Bioenergetics is the study of the ways in which food is turned into ATP. Metabolism refers to all the chemical reactions that occur in the body to maintain itself. Exercise metabolism refers to the examination of bioenergetics as it relates to the physiological changes and demands placed on the body during exercise. Metabolism dynamically adjusts to supply ATP depending on the intensity and duration of an activity.

This is done through three metabolic pathways-- the ATP-PC system, glycolysis, and the oxidative system. The ATP-PC system generates energy very quickly for high-intensity, short-duration activities. Glycolysis generates energy moderately quickly. And the oxidative system generates energy relatively slowly but for a long duration. The first law of thermodynamics states that energy cannot be destroyed or created but instead is recycled from one form into another.

In humans, this concept is referred to as energy balance. And it directly relates to body composition management. Someone that expends more than they consume will lose mass or weight. On the flip side of the coin, if more energy is consumed than expended, mass will be gained. In essence, all of that energy that someone consumes is either used or stored in the body.

The fuels for energy and exercise are mainly glucose and free fatty acids, but protein and ketone bodies can also be used under certain circumstances. Let's take a closer look at each. Glucose can be made in the body from fats and amino acids, but the majority of our daily glucose comes from eating carbohydrate-based foods. Carbohydrates are broken down into glucose through digestion. Glucose is then absorbed and transported in the blood, where it circulates until it enters cells and is either used to make ATP or stored for later.

When it's stored, it's stored as string molecules in a branched structure called glycogen. Glycogen is stored in the liver and muscle cells and can be broken down rapidly to provide energy when there is not enough free glucose in the blood. Glucose makes a relatively small contribution to overall energy production during rest or low-intensity exercise. The brain always requires glucose to function, but fats are what primarily fuel the body when it's not active.

As the intensity of an activity increases, however, the body transitions from using mostly fat as fuel to using mostly glucose to provide energy. Fats, or lipids, are the energy source that is particularly important during rest and lower-intensity activity. The chemical form in which most fats exist in food, as well as in the body, are called triglycerides, more commonly referred to as free fatty acids when they're in the bloodstream.

They're derived directly from fats and foods or are made by the body to store excess energy when more food is consumed than is needed to support daily activity. Before cells can use consumed fat or store body fat as a fuel source, it first needs to be broken down into free fatty acids. Free fatty acids are then used exclusively in the aerobic metabolic pathway to produce ATP, like the oxidative system.

One of the benefits of having fat as a fuel source is that even relatively lean people still have a large supply stored on their body, which can be broken down and used for energy during prolonged lower-intensity physical activity and exercise. The third fuel source are proteins, which are made up of amino acids. Humans use 22 different amino acids to assemble bodily proteins, nine of which are called essential amino acids.

This means that the body can't synthesize them on its own, and they have to be consumed in our diet. When a person consumes protein, it's broken down into its component amino acids. Those amino acid building blocks will then ideally be used to synthesize human bodily proteins that build up muscle and repair cellular machinery. However, the amino acids from dietary protein can also supply energy for ATP production if carb and fat sources are low. This is why adequate carb intake is important, especially after intense exercise.

Glycogen stores are replenished from the carb source, and amino acids can fulfill their main post-exercise role of building and repairing muscle. Lastly, it's important to understand the role of ketone bodies. That's the name collectively used to refer to three molecules that can be aerobically metabolized, similar to glucose. They are acetone, acetoacetic acid, and beta hydroxybutyric. And they are produced by the liver as a byproduct of the breakdown of fatty acids or through the conversion of ketogenic amino acids.

Our bodies don't have the ability to store these molecules, so they are only used acutely to produce energy and are not stored for later, like glycogen. When carbohydrate stores run low, ketone bodies are produced and are used alongside gluconeogenic glucose to help make up for the deficiency. During this metabolic state, the body is said to be in ketosis. For most people, ketones make up a small portion of the energy-producing substrates in the human body, even in nutritional ketosis.

However, there is some research studying the effect of ketosis and ketone-producing diets. So it's good to know exactly how they work. Just like a car, the body needs fuel to run. Essentially, the human body is an organic machine that can turn chemical energy into mechanical work, about 40% of which goes to cellular work, like muscle contractions, with the remainder being released as heat.

As already mentioned, there are three energy systems that generate the energy or ATP needed for muscle contraction. They're distinguished by the rate at which they produce energy and the substrates that are used. The ATP-PC system utilizes stored ATP within the muscle and can also take a phosphate from a phosphocreatine molecule and attaches it to ADP to create ATP. This process does not require oxygen, and it's considered anaerobic.

Glycolysis exclusively uses carbohydrates in the form of glucose or glycogen to produce ATP and lasts for approximately 30 to 60 seconds of duration. This process can be both anaerobic and aerobic. The oxidative system is the only energy pathway that can utilize all types of substrates, including carbs, fats, proteins, and ketone bodies. However, it produces ATP relatively slowly and requires oxygen to do so. This system is an aerobic process.

When it comes to exercise activities, they can be categorized in two ways, by intensity and duration. During a steady-state activity, the rate at which energy needs to be produced is relatively constant. Steady-state activities are also characterized as being at least several minutes in duration. By comparison, intermittent or interval activities' energy demands are constantly fluctuating throughout the activity. Periods of high-intensity work require the ATP-PC system and glycolysis to provide the ATP needed, whereas during rest periods, the oxidative pathway will contribute more to ATP production.

So what we're saying here is that during interval training, all energy systems will contribute to ATP production at some point during the workout. If the exercise duration is at least several minutes and the intensity is moderately high, the body will need to breathe more to deliver more oxygen to the mitochondria in order to support the work of oxidative phosphorylation.

This shift to a larger breathing volume is known as the first ventilatory threshold or VT1. VT1 is also the point at which carbs and fat contribute equally to energy production. However, if the exercise intensity continues to increase, the body won't be able to rely on oxidative metabolism, thus shifting to anaerobic processes that no longer use free fatty acids as a substrate and rely more on carbohydrates.

This shift in fuel utilization also means that more carbon dioxide will be produced. As you probably recall, carbon dioxide is a byproduct that needs to be exhaled. As a result, breathing rate will increase, and consistent talking will become more difficult. This is the second ventilatory threshold, or VT2, which is important not only to deliver oxygen but also to exhale the increasing amounts of carbon dioxide being produced.

So to recap, at VT1, a person needs to take in more air to supply more oxygen to the mitochondria. At VT2, the body becomes more focused on exhaling carbon dioxide. Moving on, on any given day, you'll need energy from calories for different processes. These can be broken down into your Resting Metabolic Rate or RMR, the Thermic Effect of Food or TEF, Non-Exercise Activity Thermogenesis or NEAT, and Exercise Activity Thermogenesis, which we call EAT.

The calculations work a little something like this. If a person has a higher RMR or engages in more NEAT or EAT, that person will need more calories to generate the ATP to support those activities. Makes sense, right? Some foods, like proteins, generate a higher TEF than others. To lose weight, which is often why we find ourselves talking about metabolism, it's usually more effective to consume fewer calories than to try to create the necessary energy deficit through activity alone.

That's a foundational premise for all the diets we've ever seen. It just takes a bit of science and investigation to be sure we're doing it in a holistic, balanced way. Daily energy intake through the foods we eat needs to be adequate to maintain a healthy body weight, allow for proper bodily function, and support physical activity. If daily food intake is matched to energy needs, a person is said to be in energy balance. And that's a good state to be in.

We're always thinking of better ways to keep our clients engaged and adhering to their fitness program. And by understanding the concepts within this chapter, we can offer better solutions. I mean, think about being able to say, hey, so let's try upping your NEAT by walking your dog for an extra 15 minutes per day and adding some gardening activities as well as taking the stairs at work. And then let's up your EAT through the intensity of your workouts. And finally, let's optimize TEF in your daily foods as well.

That's way better than saying so you're going to have to run 10 miles, eat one salad and a gallon of lemon turmeric water every day forever. The bottom line is, knowing the ins and outs of metabolism and energetics will make you a better fitness professional by being one that uses science and a balanced approach rather than trends that are hard to stick with and only show short-term results. As with everything else, we're looking for longterm solutions and ways to inspire our clients to live better lives. OK, I think that's good for now. Be well, and happy studying.